EXP-8 NMAP to discover live hosts using scans

- Aim:

To discover live hosts across network segments using various NMAP scanning techniques (ARP, ICMP, TCP/UDP) on TryHackme

- Theory:

- ARP scan (-PR):

Uses address resolution protocol frame, extremely fast and reliable but strictly confined to the local subnet

- ICMP scan:

-PE: ICMP echo request

-PP: ICMP timestamp request

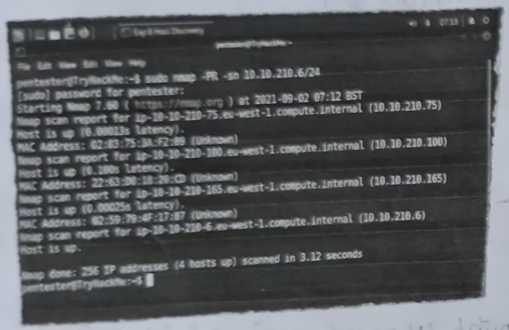
-PM: ICMP address mask

- TCP ACK ping:

-PA <ports> sends a TCP ACK packet directly. an active target responds with RST

- UDP ping:

-PU <ports> sends UDP packets to closed ports expecting an ICMP type 3 code 3 error message



- Procedure:

⑤ ① Target specification & preview:
preview scan list without sending traffic

nmap -sL -n 10.10.12.8/29

② ARD host discovery:
scan local subnet using ARP queries

sudo nmap -PR -sn 10.10.210.0/24

③ ICMP Host discovery
ICMP echo discovery: sudo nmap -PE -sn
ICMP timestamp discovery: sudo nmap -PP -sn
ICMP address mask discovery: sudo nmap -PM sn

④ TCP/UDP host discovery:
TCP SYN ping: sudo nmap -PS80
TCP ACK ping on 80: sudo nmap -PA80
UDP ping scan: sudo nmap -PU53,161

- Output
Starting Nmap 7.60 (https://nmap.org)
Nmap scan report for ip-10-10-210
Host is up (0.00013s latency)
MAC address: 02:83:75:3A:F2:89

Nmap scan report for ip - 10-10-210

Host is up.

MAC addres : 22:63:DD:1E:20:2D

Nmap done : 256 IP addddress

" " [span 44](start_span) [span-44] (end_span)

Result :

The various live hosts discovery mechanism of NMAP (ARP, ICMP, TCP & UDP ping scans) were studied and executed on TryHackme